

Thermo-osmotic coefficients in membrane distillation: Experiments and theory for three types of membranes

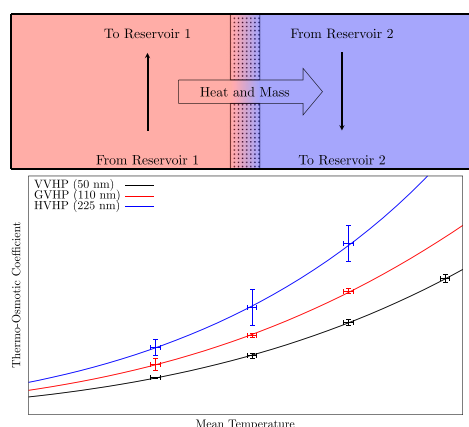
Kim R. Kristiansen, Øivind Wilhelmsen, Signe Kjelstrup*

PoreLab, Department of Chemistry, Norwegian University of Science and Technology, 7491 Trondheim, Norway

HIGHLIGHTS

- New apparatus and recommended methodology for measuring thermo-osmotic coefficients.
- Temperature-dependence of thermo-osmotic flux through membranes analyzed in detail.
- Mechanical power density from pressure-retarded membrane distillation (PRMD) discussed.
- Present limitations of PRMD analyzed. Membrane properties identified as limiting factors.
- Specific recommendations for membrane optimization for PRMD are provided.

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:

Membrane distillation
Non-equilibrium thermodynamics
Thermo-osmotic coefficient
Thermal diffusion
Pressure-retarded flow
Interface transport

ABSTRACT

The thermo-osmotic coefficient of a porous medium is the ratio of the mass flow rate and the temperature difference that drives the flow. Thermo-osmotic coefficients of the Millipore DuraPore HVHP, GVHP and VVHP membranes have been measured using a new apparatus designed to provide accurate mass flux measurements with tightly controlled temperatures, pressures, and compositions on the feed and distillate side. To properly analyze the experimental data, recommendations on data reduction procedures anchored in non-equilibrium thermodynamics are discussed. An expression is presented for the apparent energy of activation of the transport process in terms of the derivative of the thermo-osmotic coefficient with respect to the inverse mean temperature. The expression is shown to accurately predict the temperature dependence of the thermo-osmotic coefficients from experiments. The expression is next used to explain observations in similar porous systems from the literature. Based on this, an optimization scheme to find the optimal mean pore radius to maximize the power density of a pressure-retarded membrane distillation (PRMD) process is presented. For membranes with properties similar to the Durapore XXHP series membranes, the optimal pore radius is found to be between 1 and 10 nm. It is shown that the power density can benefit greatly from a narrow pore size distribution and from membrane surface treatments that increase the liquid-solid contact angle, potentially exceeding a benchmark minimum power density of 5 W/m² (set for the feasibility of pressure-retarded osmosis). An assessment of the

* Corresponding author.

E-mail addresses: kim.kristiansen@ntnu.no (K.R. Kristiansen), oivind.wilhelmsen@ntnu.no (Ø. Wilhelmsen), signe.kjelstrup@ntnu.no (S. Kjelstrup).

second law efficiency of the PRMD process is also presented. A critical part of the process is due to the inevitably smaller water flux through smaller pores, required to maintain a high pressure difference and a feasible power density. This can only be alleviated by increasing the temperature difference across the membrane. We conclude that membranes designed for the PRMD process need increased overall thermal resistance to enhance performance.

1. Introduction

Direct contact membrane distillation (DCMD) is a process where a temperature difference between two water reservoirs drives a net flux of water vapour through a hydrophobic porous medium separating the two, effectively distilling water when the hot side contains impure water and the cold side pure water. A myriad variants of the membrane distillation (MD) process have been studied in detail since the first DCMD paper by Findley was published in 1967 [1]. Measurements of water fluxes driven by a temperature difference across hydrophobic membranes have been reported for several decades, and the measurements published until 2010 are thoroughly summarized in the book by Khayet and Matsuura [2] and sources therein. In these works, the flux is usually described as the mass transfer coefficient times the difference in the vapour saturation pressure difference at the two liquid boundaries. As was shown in [3], this expression can be derived by approximating that the total heat of transfer across the membrane system is equal to the enthalpy of vaporization. Without this approximation, it is necessary to define the thermo-osmotic coefficient as an independent factor that is different from the mass transfer coefficient.

More recently, a non-isobaric concept, which we henceforth refer to as pressure-retarded membrane distillation (PRMD), was demonstrated under the concept name of MemPower [4]. It was investigated both theoretically and experimentally by several groups [5–12], using the basic theory that had been established in the MD literature for modelling and interpretation of experiments. A thorough non-equilibrium thermodynamics assessment of the process was presented by Keulen et al. [13], and several other groups have also given attention to the process [14–16]. The MemPower concept, which simultaneously produces distilled water and mechanical power by passively maintaining a pressure difference against the MD flux, is interesting because of its flexibility. It can potentially exploit low-grade waste heat to simultaneously produce potable water and high-grade mechanical energy. Recent work has pointed out the importance of the state of the liquid-membrane interface in cases where the resistances to heat and mass transfer are comparable [17].

In this work, we follow a detailed theoretical exposition of the PRMD process [3], where a model is constructed with thorough anchoring in the theory of non-equilibrium thermodynamics. The model, although more complicated than the basic theory commonly used in the MD literature, has the advantage of properly taking into account the heat-mass coupling mechanisms and the effects of interfacial transport phenomena. We make recommendations for planning of new experiments and propose data reduction procedures. The model will be applied and used to interpret experimental data. To this end, the commercial Millipore PVDF membranes HVHP, GVHP and VVHP have been studied experimentally by use of a new experimental facility with design and equipment chosen to achieve high accuracy. Relevant properties of these membranes such as thermal conductance [18], liquid-solid contact angle [19], pore size distribution [20] and DCMD fluxes under various conditions have already been reported in the literature [21–24]. This allows us to focus on the methodology and the application of the theoretical model, while at the same time comparing the measurements to values that have previously been reported in the literature. Furthermore, we address that much of the existing theoretical treatment of the MD process consider the primary driving force to be the difference in the vapour saturation pressure [25–28] across the membrane pore, while others anchor their theory in two driving forces; the thermal and the chemical

driving forces as dictated by non-equilibrium thermodynamics [29].

When the mechanisms of transport are properly understood, knowledge of the thermo-osmotic coefficient, the permeability, and the structural properties of the membrane in combination with theory can be used as a basis for optimization schemes that can guide further improvements of the membranes to maximize the performance of the PRMD process. We will discuss a scheme that takes into account the limitations set by the liquid entry pressure of the membrane, which is the pressure at which the largest pores in the membrane start to leak liquid. Such leakages undermine the process because they lead to severe loss of both the potable water product and mechanical energy stored, by creating viscous flow through the membrane back into the feed solution.

Eventually, we will calculate the entropy production of the process for an example case involving one of the membranes and use this to estimate the second law energy efficiency of the process. This analysis reveals a weakness of the PRMD process that should be rectified by increasing the thermal resistance of the membrane system.

We begin by establishing the theory and relevant equations in Section 2. The experimental rig and procedure are presented in Section 3. Theory and experiments are compared and discussed in Section 4 before concluding remarks are provided in Section 5.

2. Theory

The DCMD cross-flow system is sketched in Fig. 1, where pumps are used to make the liquid phases flow in opposite directions parallel to the membrane surfaces to promote mixing and to reduce the effects of stagnant layers next to the membrane. The theoretical analysis of this system was discussed in detail in another paper [3], which is referred to for further details. The first order expressions for the steady-state flux densities of heat and mass are [3].

$$\begin{aligned} J_q^o &= -\Lambda_{\text{tot}} \Delta_{1,2} T + q^{*o} J_w, \\ J_w &= -D_T \Delta_{1,2} T - L_p [\Delta_{1,2} p - \Pi \Delta_{1,2} m], \end{aligned} \quad (1)$$

where J_q^o is the outgoing flux density of measurable heat as indicated in Fig. 1, and the water mass flux density is $J_w = J_w^i = J_w^o$ at steady-state conditions. The heat of transfer q^{*o} is the heat transferred reversibly along with the water flux out of the membrane, and Λ_{tot} is the total thermal conductance of the membrane system. The water flux depends on the temperature difference $\Delta_{1,2} T$, the molality difference $\Delta_{1,2} m_s$, and the pressure difference $\Delta_{1,2} p$ by the transport coefficients L_p and D_T , and the molal osmotic pressure coefficient Π . Neglecting the effects of pore size variance and interfacial mass transfer resistance, the thermo-

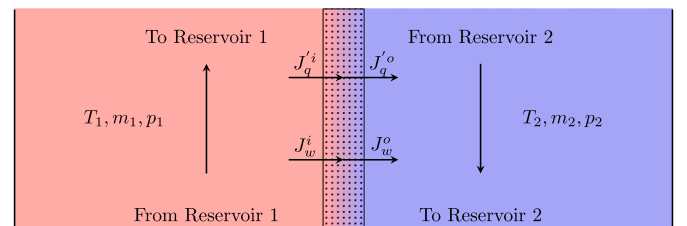


Fig. 1. Sketch of a cross-flow direct-contact membrane distillation setup, where the membrane system is treated as a black box with incoming and outgoing fluxes of heat and mass. The vertical arrows indicate directions of the bulk flow that circulates the solutions between the reservoirs and the membrane system.

osmotic coefficient can be expressed as [3]

$$D_T = - \left(\frac{J_w}{\Delta_{1,2}T} \right)_{\Delta\mu_{T,w}=0} = \frac{\Theta \phi M_w \hat{p}_w D_p(\bar{a})}{R^2 \bar{T}^3 \tau_p \delta} (q_m^* + \Delta_{\text{vap}} H_w). \quad (2)$$

Here, $\bar{T}$ is the mean temperature, R the universal gas constant, $\hat{p}_w$ the geometric mean partial pressure of the water vapour, and D_p the pore diffusion coefficient. The membrane parameters involved are the porosity ϕ , the mean pore tortuosity τ_p , the membrane thickness δ , and the mean pore radius $\bar{a}$. The factor Θ accounts for thermal polarization, as derived in [3],

$$\Theta = 1 - \Lambda_{\text{tot}} \left(\frac{\delta_1^{\text{pol}}}{\lambda_1^w} + \frac{\delta_2^{\text{pol}}}{\lambda_2^w} \right) - D_T \left(\frac{q_1^{*,w} \delta_1^{\text{pol}}}{\lambda_1^w} + \frac{q_2^{*,w} \delta_2^{\text{pol}}}{\lambda_2^w} \right), \quad (3)$$

where λ_i^w and $q_i^{*,w}$ are the thermal conductivity and heat of transfer in the liquid phases, and δ_i^{pol} the effective polarization layer thickness. Subscripts 1 and 2 denote the boundary layers on the two different sides of the membrane. The boundary layer thicknesses δ_i depend on the flow conditions near the membrane surface, where promoting turbulent flow leads to smaller δ_i and therefore smaller thermal resistance in the polarization layers. In the limit as these layers are eliminated, $\Theta \rightarrow 1$. The effective heat of transfer, $q^{*,o}$, of the system is split into two terms [3]: the enthalpy of vapourization $\Delta_{\text{vap}} H_w$, and a remainder q_m^* that accounts for the kinetic heat-mass coupling in the membrane pores. The latter is commonly neglected due to typically being two orders of magnitude smaller than the former. The operational definition of the coefficient (the equality to the left in the equation) applies to one or two-phase flow. In the absence of phase transitions, the name thermal diffusion coefficient is more common [30]. We shall use the name thermo-osmotic coefficient to indicate that a pressure difference may be involved as a driving force. Detailed discussions on this can be found in Refs. [3, 17].

The geometric mean partial pressure $\hat{p}_w$ is assumed to be proportional to the saturation pressure at the vapour-liquid interfaces, which depends exponentially on the inverse temperature of the liquid phase. This means that D_T depends roughly exponentially on the inverse temperature, giving a nearly straight line in an Arrhenius plot. Earlier experimental results that show such behavior have motivated this assumption [17]. It follows that we can define an effective energy of activation, E_a for the transport process. The first approximation to this energy of activation is found by taking the appropriate derivative of Eq. (2)

$$\begin{aligned} E_a - \Delta_{\text{vap}} H_w &\equiv -R \left(\frac{\partial \ln D_T}{\partial 1/\bar{T}} \right)_{p,m} - \Delta_{\text{vap}} H_w \\ &= -R \bar{T} \left[2 + \frac{\Delta_{\text{vap}} H_w - \bar{T} \Delta_{\text{vap}} c_{p,w}}{q_m^* + \Delta_{\text{vap}} H_w} - \left(\frac{\partial \ln D_p}{\partial \ln \bar{T}} \right)_{p,m} - \left(\frac{\partial \ln \Theta}{\partial \ln \bar{T}} \right)_{p,m} \right], \end{aligned} \quad (4)$$

where $\Delta_{\text{vap}} c_{p,w}$ is the difference in isobaric heat capacity between the vapour and the liquid phase. Since this was obtained from the first approximation to D_T in Eq. (2), this is a first approximation to E_a . In order to arrive at Eq. (2), effects from pore size variation, interfacial transport resistance, and nonlinear effects are neglected. The expression including such effects is far more complex [3], and the form in Eq. (2) is commonly encountered in the membrane distillation literature where the aforementioned effects are negligible. We shall see that an Arrhenius plot of the thermo-osmotic coefficient will give an apparent activation energy of the order of the enthalpy of vapourization, but with different magnitude due to the right-hand side of Eq. (4). Explicit expressions for D_p and q_m^* can be found in [3]. We refer to Ref. [3] for detailed expositions of all the transport coefficients involved and a numerical method for predicting them even beyond the linear regime. Reporting the values of D_T and E_a at a specified temperature yields a simple predictive formula for other temperatures

$$D_T(T) = D_T(T_0) \exp \left(- \frac{E_a}{R} \left(\frac{1}{T} - \frac{1}{T_0} \right) \right). \quad (5)$$

We will also discuss the performance of the non-isobaric, pressure-retarded MD process, where water is distilled against a mechanical pressure difference in order to simultaneously produce distilled water and mechanical power. The entropy production of the system, which is proportional to the energy that is dissipated during the process was presented in [3]. Formulated in terms of the driving forces and the transport coefficients, it becomes

$$\begin{aligned} \sigma &= \left(\frac{\Lambda_{\text{tot}}}{T_1 T_2} + \frac{V_w D_T^2}{T_1 L_p} \right) (\Delta_{1,2} T)^2 + \frac{L_p V_w}{T_1} (\Delta_{1,2} p - \Pi \Delta_{1,2} m)^2 \\ &\quad + \frac{2 D_T V_w}{T_1} \Delta_{1,2} T (\Delta_{1,2} p - \Pi \Delta_{1,2} m). \end{aligned} \quad (6)$$

The first two terms on the right-hand-side are always positive, and represent the entropy production due to heat conduction across the membrane, and due to back diffusion of water towards the feed solution. The third term, however, may take either sign, depending on the alignment of the thermal and chemical driving forces. In the MD process, $\Delta_{1,2} m > 0$ when $\Delta_{1,2} T > 0$, the third term is negative and contributes to a reduction in the entropy production, because work is recovered from the thermal energy in the form of separation of pure water from the salt solution. Furthermore, adding a pressure difference $\Delta_{1,2} p < 0$, that is, the pressure of the permeate is greater than that of the feed, even more work is recovered in the form of mechanical energy. At the same time, this also increases the magnitude of the second term, because of the increased driving force for back diffusion of water vapour towards the feed solution, due to $L_p > 0$. The entropy production is at a minimum when the applied pressure difference satisfies

$$\Delta_{1,2} p_{\min} = \Pi \Delta_{1,2} m - \frac{q_m^* + \Delta_{\text{vap}} H_w}{V_w T_2} \Delta_{1,2} T. \quad (7)$$

The reason why this is a minimum is that it represents the exact balance of driving forces such that $J_w = 0$, as can be seen by inserting Eq. (7) into Eq. (1), which means that both the power production rate and water distillation rate are zero. A more attractive objective is to maximize the mechanical power density w_{mech}

$$w_{\text{mech}} = \frac{J_w \Delta_{1,2} p}{\rho_w}, \quad (8)$$

where ρ_w is the mass density of water. The applied pressure difference that maximizes w_{mech} is

$$\Delta_{1,2} p_{\text{opt}} = \frac{\Pi}{2} \Delta_{1,2} m - \frac{q_m^* + \Delta_{\text{vap}} H_w}{2 V_w T_2} \Delta_{1,2} T = \frac{1}{2} \Delta_{1,2} p_{\min}. \quad (9)$$

This optimum is nearly universal for all membranes, due to the weak influence of q_m^* compared to $\Delta_{\text{vap}} H_w$ [3]; the second term on the right-hand side varies between 40 bar per kelvin at 298 K and 30 bar per kelvin at 363 K. The optimal pressure difference for power generation is the one that provides half the water flux that the isobaric system gives. Increasing the pressure difference beyond this point lowers the power because the resulting reduction in the water flux outweighs the additional mechanical energy obtained from the remaining flux against the pressure difference. The pressure-retarded MD process ideally operates with a pressure difference between zero and $\Delta_{1,2} p_{\text{opt}}$, adjusting the pressure difference according to the demand for water.

As discussed in the preceding paragraph, the use of a passive pressure difference is limited by the membrane's capacity to exert a large enough capillary pressure to prevent intrusion of liquid water and maintain liquid-vapour interfaces. Intrusion of liquid water leads to short-circuiting the membrane by opening channels through which viscous flow occurs, driving a water flow opposite the thermally driven flow and severely hampering the PRMD process. The maximum pressure that a

membrane can withstand before leakage of liquid occurs is referred to as the liquid entry pressure [31]. The simplest approach to obtain an idea about its magnitude is by use of the Young-Laplace equation for an interface in a solid cylinder

$$p_{LE} = -\frac{2\cos(\theta)\gamma_w}{a_{\max}}, \quad (10)$$

where θ is the solid-liquid contact angle, γ_w the water-air surface tension, and $a_{\max}$ the practical maximum pore radius. We see that this depends both on the bulk structural property $a_{\max}$, and the surface property θ . Since θ is determined by the membrane characteristics, an interesting optimization problem emerges if we let

$$\Delta_{1,2}P = -\min(p_{LE}, -\Delta_{1,2}P_{opt}). \quad (11)$$

That is, we want to operate at the optimal pressure difference, but are restricted to the liquid entry pressure of the membrane if it is smaller. Thus the practical optimum applied pressure difference is dictated entirely by the process conditions and the minimum capillary pressure of the membrane. We can then determine the optimal pore size that maximizes power generation by determining the maximum of $J_w \Delta_{1,2}P$ with respect to the maximum pore radius. Results of such an optimization will be discussed in Section 4.

3. Experimental

A dedicated experimental apparatus designed and built for high accuracy was assembled as part of this study. We will next describe this apparatus and the procedures used to study the thermo-osmotic coefficient.

3.1. Cell

The cell unit is illustrated in Fig. 2. The cell design is inspired by the work of Gerstandt et al. [32], and features a simple design where a narrow flow channel is cut into two blocks of polyetheretherketone (PEEK) framed by slots for O-rings. The flow channels feature holes designed to fit 1/16" tubing to bring thermocouples into the flow channels. The membrane system is then formed by sandwiching a membrane between the outer O-rings and holding it all together by an array of nuts and bolts. The cell features inlet and outlet ports designed to be on the top of the cell, in order to guarantee complete flooding of the flow channel. The materials are chosen to withstand the corrosive environment of saline water at elevated temperatures, and at the same time handle pressures up to at least 30 bar.

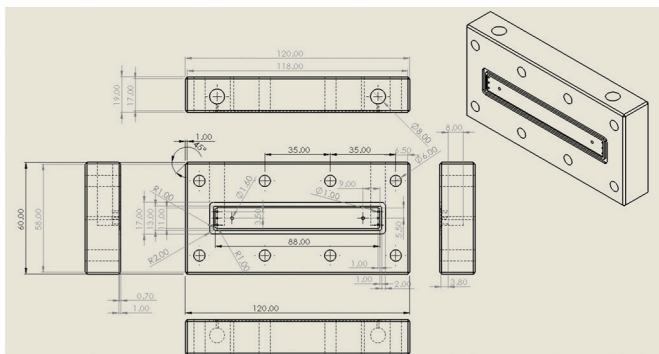


Fig. 2. PEEK half-cell used for the thermo-osmotic measurements. Two such half-cells are designed to accommodate a membrane sandwiched between two O-rings encircled by the flow channels. Inlets and outlets are drilled at the top of the cell to guarantee complete channel flooding, and holes are drilled to accommodate thermocouples directly into the flow channels.

3.2. Membranes

The membranes used in this work are the Millipore commercial flat sheet Durapore polyvinylidene fluoride (PVDF) membranes, VVHP, GVHP, and HVHP. The structural properties provided by the manufacturer are listed in Table 1. These membranes were chosen because the GVHP and HVHP membranes are already well-studied in the literature, such that we may focus on the influence of changing the process conditions. Furthermore, all of the membranes are made from the same material and have similar properties except for their pore sizes. This allows us to properly assess the effect of pore size on the process performance. The average solid-liquid contact angle is 111° [19].

3.3. Flow system

The apparatus sketch in Fig. 3 shows the flow system as two closed loops where the flow is driven by two Hydra-Cell MT8 positive displacement pumps. Two insulated reservoirs were placed on top of Ohaus Explorer Ex2202 electronic weights, denoted W_1 and W_2 in the figure, at an elevated position relative to the pumps. The solutions were pumped through heat exchange baths where tubing was coiled up and immersed in silicon oil which temperature was regulated within 0.2 K by means of two Grant LT Ecocool 150 thermostats. The solutions would then proceed through a series of valves V_1 - V_5 , designed to allow measurement of flow rate through both flow loops by means of a Bronkhorst ES-1131-AGD ultrasonic flowmeter, denoted F . The valve V_5 is connected to a sink to allow the flushing of one fluid through the valve system to clean it before reconnecting it to the membrane cell. The pressure at the cell inlets and outlets are measured by means of Swagelok MTO S-model transducers, denoted P_{ij} , and MTO pressure regulators R_1 and R_2 control the pressures. Temperatures were measured at two positions in each flow channel by means of Type E thermocouples, denoted T_{ij} . The scales, thermocouples, flowmeter and pressure transducers all reported their output to an Agilent 34970 A data logger, controlled by a local computer via a custom LabView program. The pumps were controlled by the same computer by means of ABB ACQ580 frequency transformers, giving full control of the pumping rates. The Grant LT Ecocool thermostats were controlled by the same computer using the LabWise software supplied by the manufacturer. All tubes and metal parts in contact with the fluid in one of the flow loops were constructed by Hastelloy C or Monel alloys in order to withstand corrosion. The other flow loop was made of 316 stainless steel, designed for distilled water only. Both flow loops were designed for a pressure of at least 30 bar over extended periods of time. All tubings were thermally insulated by covering them in foam tape and an external layer of fiberglass cloth.

3.4. Solutions

The solutions were prepared by adding pure reagent grade NaCl to a weighed amount of distilled water (conductivity < 1.5 $\mu\text{S}/\text{cm}$). Solutions were prepared with 0.1, 0.3, 0.5, 0.7, and 1.0 mol/kg NaCl/ H_2O . The conductivity of all solutions was measured with a Phillips PW 9529 conductivity meter before and after each experiment, in order to verify that the salt concentration had not changed in a way that would indicate leakage of salt across the membrane system. We use molality or salinity as composition measure as opposed to molarity, because the

Table 1

Properties of hydrophobic PVDF membranes provided by the manufacturer Millipore: mean pore radius $\bar{a}$ and porosity ϕ . The thickness δ was measured by means of a micrometer on-site.

Membrane	$\bar{a}$ (nm)	ϕ	δ (μm)
HVHP	225 $\pm$ 1	0.75 $\pm$ 0.01	100 $\pm$ 3
GVHP	110 $\pm$ 1	0.75 $\pm$ 0.01	122 $\pm$ 2
VVHP	50 $\pm$ 1	0.70 $\pm$ 0.01	92 $\pm$ 4

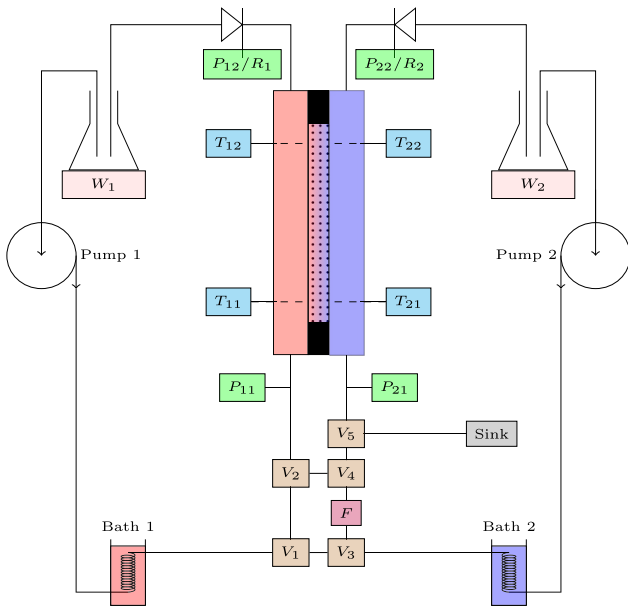


Fig. 3. Sketch of the apparatus used for thermo-osmotic measurements. The membrane is sandwiched between two half-cells with flow channels that allow for flow of liquids tangentially to the membrane surface, by means of pumps circulating the liquids from their reservoirs. Pressures and temperatures are recorded by the pressure transducers P_{ij} and the thermocouples T_{ij} , and reservoir weights are continuously monitored by electronic scales W_i . Thermostatted baths and pressure regulators R_i allow for control of the pressure and temperature in each flow channel separately, and a flowmeter F records the flow rate through one of the channels at any time. A series of valves V_i is installed for calibration purposes.

temperatures change over the course of the experiments. This affects the fluid density and therefore changes the molarity, whereas the molality or salinity is fully independent of temperature.

3.5. Measurement procedures

Each measurement series with a new solution on the saline side started with purging the old solution from the system, and then allowing the flow loop to run for 1 h with the new solution, before refreshing that solution to minimize dilution effects that may have occurred in the initial run. The pump frequency was calibrated against the flow-meter at the set flow loop pressure, in order to ensure equal flow rates in both flow loops. The flow system was allowed to run with both flow loops active at isothermal conditions for 3–6 h until the reservoir weights fully stabilized. The joints and other weak spots were checked with a dry paper towel for any leakages, and the total liquid mass was recorded over time to detect any minor leakages. The flow rates were set just above 100 mL/min, giving an estimated average flow velocity of 15 cm/s tangentially to the membrane surface inside the flow channels of the cell. With a hydraulic diameter of 1.83 mm according to flow channel dimensions shown in Fig. 2, this gives a Reynolds number in the range 200–1000 between the lowest and highest temperatures in these experiments, where flow can be assumed to be laminar. After the initial isothermal stationary state was confirmed, the temperatures in the heat baths were adjusted until both the thermocouples inside both the flow channels reported the target temperatures within 0.5 K. After both flow channel temperatures were stabilized at the target temperatures, changes in reservoir weight were monitored over time. An initial transient period followed the adjustment of temperatures due to the temperature-dependence of the fluid density. The reason for this transient period is that it takes time for the steady-state temperatures to settle throughout the apparatus, and the liquid density changes accordingly. As the density of the liquid in the piping of the apparatus

changed, more or less total mass of liquid was contained in the piping volume, thus continuously changing the liquid mass left in the reservoirs. After this initial period, the system was allowed to stabilize in a steady state characterized by a steady increase/decrease in reservoir weight due to thermo-osmotic flux across the membrane. This steady state was measured for at least 2 h before the temperatures were readjusted to different target temperatures.

3.6. Case studies

The measurement series included temperature differences of -20 K, -10 K, 0 K, 10 K and 20 K. Attention was made to keep the arithmetic mean temperature constant in each series. Measurements were repeated for different mean temperatures of 298 K, 308 K, 318 K, and also 328 K for the VVHP membrane, which proved to be the most stable of the three. For instance, for the case with $\bar{T} = 298$ K, the steady state measured at $\Delta_{1,2}T = -20$ K means that flow loop 1 was maintained at 308 K and flow loop 2 at 288 K, whereas for the one at $\Delta_{1,2}T = 10$ K, flow loop 1 was maintained at 293 K and flow loop 2 at 303 K.

For the VVHP membrane only, this procedure was repeated with NaCl concentrations of 0.1, 0.3, 0.5, 0.7, and 1.0 mol/kg in Reservoir 2, always keeping distilled water in Reservoir 1. This means that for any set with a mean temperature and composition, 5 steady-state measurements were recorded, giving a total of 25 steady-state measurements to cover the range of mean temperatures at any given solution composition.

3.7. Data analysis and reduction procedures

The time evolution of the recorded reservoir weights W_1 and W_2 give the total flux of water from one reservoir into the other by means of transfer across the membrane. This method of measuring the flux was previously demonstrated by Thorsen and Holt [33]. The water mass flux density J_w was obtained from the raw data by evaluating the mass balance of water:

$$J_w = \frac{1}{A} \frac{dW_2}{dt} = -\frac{1}{A} \frac{dW_1}{dt}, \quad (12)$$

where A is the exposed membrane area, and t is the time. The latter equality is always checked as a test of leakage in the system. If this equality holds within statistical significance, then losses from leakage are negligible. We recommend this way of measuring the water flux for two reasons: the mass of water is independent of the temperature of the reservoir, whereas the density is not. Measuring the time evolution of the reservoir weights rather than that of the volume of water in the reservoir eliminates measurement errors due to changes in density with temperature. Furthermore, checking that the two weights change at equal rates at steady state eliminates errors due to leakage in the flow system. The thermo-osmotic coefficient is then obtained by simple linear regression. In practice, it was calculated by

$$D_T = - \left(\frac{dJ_w}{d\Delta_{1,2}T} \right)_{\bar{T}, \Delta_{1,2}\mu_{T,w}}, \quad (13)$$

which is the main quantity of interest in this work. This means that D_T is obtained as the slope of J_w with respect to a change in $-\Delta_{1,2}T$ when the experiments are carried out at the same mean temperature $\bar{T}$ and the same chemical driving force $\Delta_{1,2}\mu_{T,w}$. The latter means in practice that we are changing only the temperature difference while keeping the differences in composition and pressure the same throughout all experiments undertaken to obtain a single value of D_T . We recommend this as a more robust way of obtaining D_T than reporting the value $-J_w/\Delta_{1,2}T$ at any given steady state, because it eliminates the influence of the chemical driving force on J_w , and any other effects that may otherwise remain unchecked, on the calculation of D_T .

4. Results and discussion

4.1. Discussing the measurements

Fig. 4 presents an example of measurements from the experimental apparatus (top) and the resulting water flux density (bottom). In the top subfigure, we show the recorded reservoir weights (both set to zero at the beginning of recording), as they evolve in time. In this particular case, the water provided by Reservoir 2 was maintained at roughly 308 K, and that provided by Reservoir 1 at 288 K. The weight of Reservoir 2 was steadily decreasing at the same rate at which the weight of Reservoir 1 was steadily increasing. In this example, water migrates from the hot to the cold side. The water flux density was calculated by use of Eq. (12) at several temperature differences while keeping the mean temperature of the system constant. In the following, we will use the sign convention that a positive flux means net flow from Reservoir 1 to Reservoir 2.

4.2. The overall thermo-osmotic coefficient

The thermo-osmotic coefficient was obtained for a range of mean temperatures between 298 K and 328 K for the different membranes using distilled water in both reservoirs. The obtained values are reported in Table 2 and plotted in Fig. 5, where they are compared to values predicted by the comprehensive non-equilibrium thermodynamics (NET) procedure presented in Ref. [3]. The porosities provided by the manufacturer were used in the model, and the pore tortuosity was in all cases set equal to

$$\tau_p(\phi) = \sqrt{\sum_{n=1, \text{odd}}^{\infty} \frac{(-1)^{(n-1)/2}}{n} \left(\phi^{(n-1)^2/4} + \phi^{(n+1)^2/4} \right)} \quad (14)$$

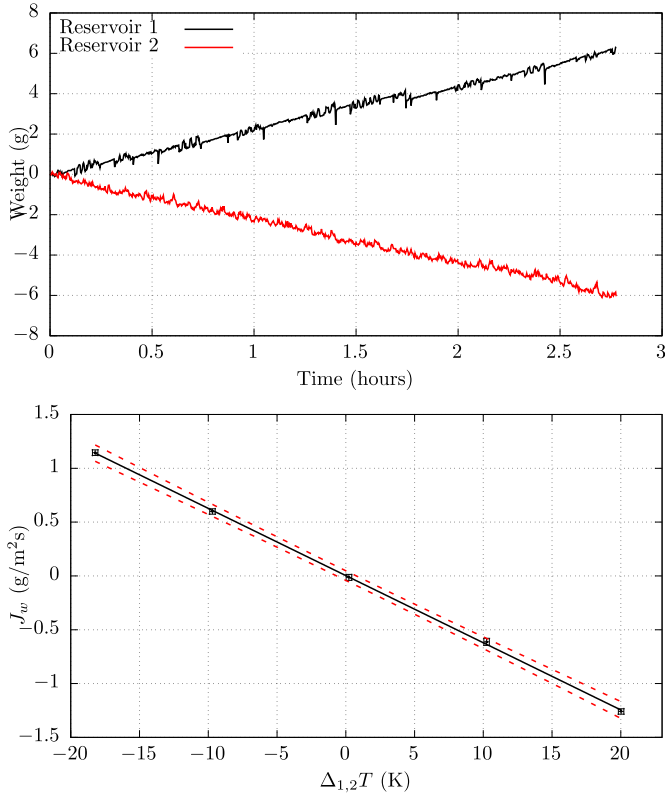


Fig. 4. Results from measurements with the Durapore VVHP membrane at a mean temperature of 298 K. Top: Time evolution of the recorded reservoir weights at steady state using distilled water with a temperature difference of approximately 20 K. Bottom: Steady state water flux density J_w as a function of the applied temperature difference $\Delta_{1,2}T$.

Table 2

Thermo-osmotic coefficients (in units mg/K s m²) from different membranes and mean temperatures.

	298 K	308 K	318 K	328 K
VVHP	62 ± 1	99 ± 4	155 ± 5	228 ± 7
GVHP	84 ± 10	133 ± 3	207 ± 4	–
HVHP	113 ± 13	180 ± 30	287 ± 30	–

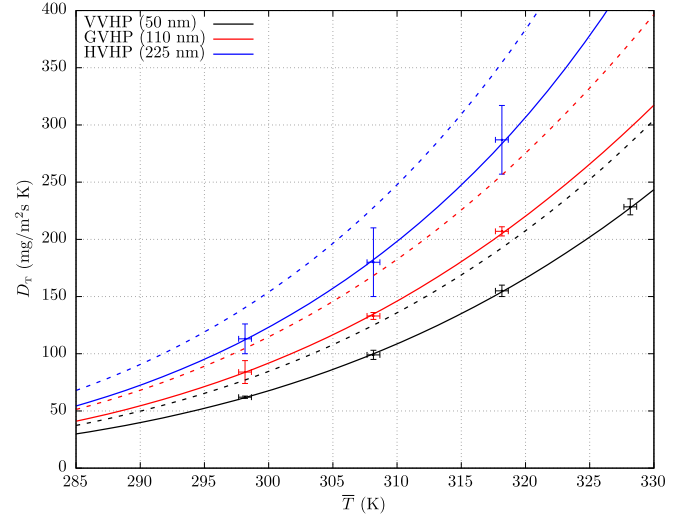


Fig. 5. Thermo-osmotic coefficients from different membranes at various mean temperatures. The solid lines represent values predicted by the NET procedure from [3], adjusted for temperature polarization, and the dashed lines give the predicted values when temperature polarization is neglected.

in accordance with the cylindrical fibre model [34]. For the investigated membranes, τ_p then ranges from 1.43 to 1.46. These values are smaller than those typically reported in most literature, but agree better with the values reported by Alves and Coelho [35].

In the literature, it is typically not D_T , but rather the flux density J_w that is reported together with the process conditions at steady-state. For example, Phattaranawik and Jiraratananon reported a water flux density of 9 g/m²s through the GVHP membrane with distilled water at temperatures 343 K and 293 K [21]. In terms of the variables used in this work, $\Delta_{1,2}T = 50$ K and $\bar{T} = 318$ K. By use of linear extrapolation from the experiments reported in this work, we estimate a water flux density of $J_w \approx D_T(\bar{T})\Delta_{1,2}T = 10.3 \pm 0.2$ g/m²s. No error bars were reported, but it is likely that the 10 % difference between these values can be accounted for by different degrees of temperature polarization.

The thermo-osmotic coefficient is positive in all cases, which is typical for hydrophobic membranes [30] and can be ascribed to dominance of the heat pipe effect, which was called phase-change-induced flow by Kim and Mench [36]. This is represented by the $\Delta_{\text{vap}}H_w$ term in the system's heat of transfer. This is the heat that is carried across the system as endothermic evaporation occurs on the feed side while at the same time exothermic condensation occurs on the permeate side. The remainder of the heat of transfer, q_m^* , is negative but much smaller in magnitude than $\Delta_{\text{vap}}H_w$, which is the primary justification for neglecting this term and leaving only the $\Delta_{\text{vap}}H_w$ term. This leads to the ubiquitous formula $J_w = -B_w\Delta p_w^*$ via the Clausius-Clapeyron relation [3], which is often used as the starting point in the MD literature [2]. However, as discussed in detail in previous work [3], we emphasize that it represents an approximation. This approximation has lead authors to cite the difference in vapour saturation pressure as the primary driving force in MD [25–28], whereas the true driving force is the difference in inverse temperature [17]. The approximation appears to be adequate if the

statistical error in the measurements is 5–10 % or greater. More accurate experiments merit to include q_m^* in the theoretical analysis.

We expect from kinetic arguments that q_m^* in the case of a liquid-filled pore is still negative, as the primary effect will be the absorption/desorption or transport of tangential momentum along the pore walls, an effect discussed in detail in [37]. In such a system where there is no phase transition to carry large amounts of heat from high to low temperature, we expect the kinetic effects to be relatively larger and possibly lead to a negative thermo-osmotic coefficient. A negative D_T means a thermo-osmotic flux directed from the cold side to the hot side. This was indeed observed by Kim and Mench [36] and others see [30] for hydrophilic membranes. A negative enthalpy of adsorption may be part of the reason for this.

4.2.1. The apparent energy of activation

Fig. 6 shows the variation of the logarithm of D_T as function of the inverse mean temperature. The sensitivity of D_T to the mean temperature can then be quantified by the apparent energy of activation, obtained from the slope in Fig. 6 and described theoretically by Eq. (4). Within the statistical uncertainty, it can be seen that all the measured values fall on the same nearly straight curve. This means that no significant difference can be detected in the apparent energy of activation for water transport through the different membranes. The three curves in the plot show predictions using different methods. The simplest but least accurate method is obtained from Eq. (5), using a constant value of E_a . This can be refined by accounting for the temperature dependence of E_a by means of Eq. (4). The dotted line in Fig. 6 is given by a temperature-dependent E_a omitting the temperature polarization term, which does not deviate much from the solid line representing the use of all terms in Eq. (4). The former approximation works well because the polarization does not change much with the mean temperature. These changes are probably from the effect of temperature on the liquid viscosity, making the polarization layers more easily disturbed at higher temperatures [3].

The formula in Eq. (4) can be used to explain why E_a is essentially the same for all the membranes at similar flow conditions. The evaporation enthalpy is a fluid property of water, and the first bracketed term on the right-hand side of Eq. (4) is the same for all membranes. The

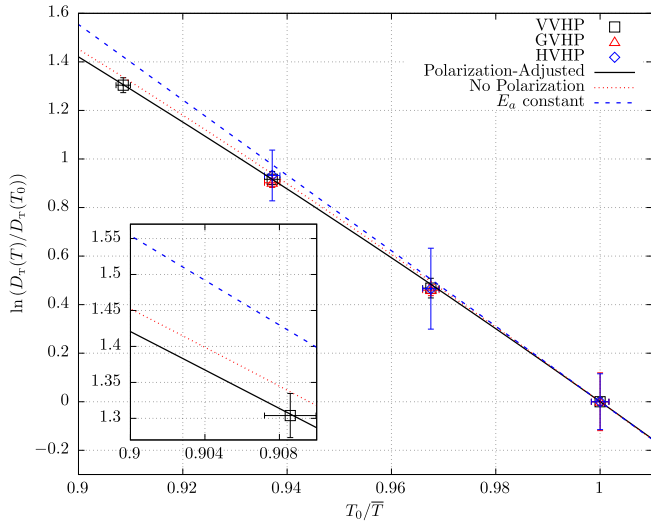


Fig. 6. Logarithm of the measured thermo-osmotic coefficient of the Millipore VVHP, GVHP and HVHP membranes as functions of the inverse mean temperature. The reference temperature $T_0 = 298\text{K}$ was chosen to obtain a dimensionless plot. The solid line indicates values predicted by Eq. (4), while the dotted line is predicted by the same equation with the temperature polarization correction excluded. The dashed line shows values predicted when using a constant energy of activation.

denominator in the second bracketed term is dominated by $\Delta_{\text{vap}}H_w$, and the fourth term depends mostly on the flow conditions rather than the membrane. This leaves only the third bracketed term as the cause of a possible difference between membranes. In the most extreme cases, $D_p \propto \sqrt{T}$ in very small pores (Knudsen diffusivity), and $D_p \propto T^{2.022}$ in very large pores (water-air interdiffusion). The third term thus varies between 0.5 and 2.022 as a function of pore size, accounting for a maximal difference of 1.522 times RT , which is roughly 4 kJ/mol. In our cases, the third term is predicted by the model to be 0.94, 1.14, and 1.27 at 298 K for the VVHP, GVHP and HVHP membranes, respectively, which gives a difference of roughly 800 J/mol difference in E_a between the VVHP and HVHP membranes. The statistical uncertainty is roughly 1 kJ/mol, making the predicted difference undetectable in the experiments. We report a common mean value of $E_a = 36 \pm 1\text{kJ/mol}$, obtained from the average slope between the data points plotted in Fig. 6. This is not significantly different from the value $35 \pm 1\text{kJ/mol}$ reported by Kerem et al. [38] for the water transfer process through hydrophobic gas diffusion layers with pore radii up to 1.5 μm . Kerem et al. performed their measurements with pure water, and did not report any phase transition. We predict the value of the fourth term on the right-hand-side of Eq. (4) to be 1.46 at 298 K, which accounts for <500 J/mol difference in E_a between the most extreme gas diffusion layers and the HVHP membrane.

The analysis in the preceding paragraph is valid under the assumption that the hydrophobic layer maintains vapour-liquid interfaces through which water must pass by phase transition in order to proceed through the material. The observations in [38] that D_T tends to be large and positive through hydrophobic layers and small and negative through hydrophilic membranes could indicate that transport in the hydrophobic case is dominated by the presence of phase transitions, at least in parts of the system. For the liquid-filled pores, which is certainly the dominant case in hydrophilic layers, the thermo-osmotic coefficient is expected to be small and negative due to diffuse scattering of molecules near the walls – since molecules from the hot side transfer on average greater tangential momentum to the wall than those from the cold side. The result is a statistical net force acting on the fluid towards the hot side. Since the thermo-osmotic coefficient of the liquid-filled case is much smaller than that in the two-phase case, a situation with a partially wet hydrophobic layer can still have a thermo-osmotic coefficient that is dominated by the dry pores where phase transitions occur. This is an explanation for why the value for E_a from Kerem et al. [38] is similar to that found in this work, even with hydrophobic GDLs that are wet. Kerem and colleagues also observed that E_a in the hydrophilic layers is smaller, but of the same order of magnitude as in the hydrophobic cases. This can at least partially be explained by the exponential dependence of the liquid water self-diffusion coefficient on temperature, which gives an apparent energy of activation between 16 and 23 kJ/mol [39].

An important feature of Eq. (4) is that E_a is independent of the structural parameters ϕ and τ_p , and only the size of the pores is needed to calculate the curves in Fig. 6. Further dependence on pore size is expected when the pores approach the nano-scale, due to a lowering of $\Delta_{\text{vap}}H_w$ when the confinement begins to influence the number of nearest neighbours for each water molecule on the liquid side of the vapour-liquid interface [40]. Since the experimental data fit the curve that is predicted by the model within the statistical uncertainty of the experiments, we conclude that no such lowering of $\Delta_{\text{vap}}H_w$ could be detected, and all the temperature dependence can be explained on the basis of Eq. (4).

4.2.2. Measurements with sodium chloride solutions

With the VVHP membrane, the D_T measurements were repeated with varying quantities of sodium chloride added to one of the reservoirs, varying the molality systematically between 0 and 1 mol/kg. As discussed in [3], the effect of solutes in the liquid phases on D_T is expected

to be primarily due to the influence on the water saturation pressure. The Gibbs-Duhem relation for the solution has been used to obtain the following relation:

$$\left(\frac{\partial \ln p_{w,\text{sat}}}{\partial m}\right) T = -\nu M_w \Gamma_s, \quad (15)$$

where Γ_s is the thermodynamic factor and ν is the van't Hoff factor of the solute, which equals 2 for NaCl. For aqueous NaCl, Γ_s is practically constant in the region of interest and equal to 0.93, based on values of the activity coefficient in the range 0–1 mol/kg [41]. The geometric mean saturation pressure when $m_1 = 0$ and $m_2 > 0$ is

$$\begin{aligned} \hat{p}_{w,\text{sat}} &\approx \sqrt{p_{w,\text{sat}}^*(T_1) p_{w,\text{sat}}^*(T_2) \exp(-2m_2 M_w \Gamma_s)}, \\ &\approx p_{w,\text{sat}}^*(\bar{T}) \exp(-m_2 M_w \Gamma_s), \end{aligned} \quad (16)$$

where $p_{w,\text{sat}}^*$ is the saturation pressure of pure water. The maximum deviation of the saturation pressure from that of pure water in the concentration range of interest is $1 - \exp(-M_w \Gamma_s)$, roughly 2 %. Furthermore, the composition difference provides a flux contribution $L_p \Pi \Delta_{1,2} m$ according to Eq. (1). Hence, we expected to see no significant change in the slope of J_w as a function of $\Delta_{1,2} T$ with varying NaCl concentration. However, we expect that the concentration difference will shift the curve in Fig. 4 upwards, and that the concentration difference is numerically equal to the intercept value at $\Delta_{1,2} T = 0$. The prefactor to the molality difference is, to first approximation

$$L_p \Pi = \frac{2M_w \Gamma_s \phi \hat{p}_w D_p(\bar{a})}{RT \tau_p \delta}. \quad (17)$$

This gives an estimate of the resulting flux density with the VVHP membrane on the order of 100 mg/s m² per unit molality difference, which is the same order as the statistical uncertainty in the intercept in these experiments. As a result, no influence from the salt concentration could be detected, and the experiments gave plots similar to the example in Fig. 4 at all salt concentrations.

The influence of the solute concentration is in the MD literature usually attributed to two effects: by a change in the water activity of the feed solution, captured here by Eq. (16), and the flux due to the chemical driving force, which is typically described as a “threshold temperature difference” that must be overcome for the process to work. In terms of the variables used in this work, the threshold is

$$\Delta_{1,2} T_{\min} = \frac{2RT^2 M_w \Gamma_s}{q_m^* + \Delta_{\text{vap}} H_w} \Delta_{1,2} m, \quad (18)$$

which is nearly universal because of the weak influence of membrane parameters on the value of the denominator, and generally smaller than 1 K per unit molality difference. Systematic studies on the effect of salt concentration are few. We note the interesting results of Godino et al. [42], who demonstrated that at elevated salt concentrations, severe thermal polarization may invert the direction of the water flux as the chemical driving force begins to dominate.

4.3. The pressure-retarded process

4.3.1. Performance

Since the molal osmotic pressure coefficient Π is a thermodynamic equilibrium property of the salt solution that does not depend on the properties of the membrane, a measurement of J_w resulting from a difference in solute composition is equivalent to measuring the flux due to a difference in pressure, as the resulting flux is in both cases dictated by their common transport coefficient L_p . The presence of salt does not alter the mechanism of transport in the membrane. A molality difference of 1 mol/kg provides a chemical driving force equivalent to a pressure difference of roughly 50 bar, which is far beyond pressure levels that may realistically be maintained without penetration of liquid water into the

pores of membranes presently in use. Provided that the pressure is kept below the liquid entry pressure of the membrane, passively maintaining an elevated pressure on the permeate side allows the extraction of mechanical power from the pressurized permeate, with a negligible loss of distillation yield. The practical issues that presently hamper development of this technique relate to the mechanical properties of the membranes: they must be able to withstand large pressure differences without rupturing and without intrusion of a liquid-phase into the pores.

For the membranes studied in this work, the bubble points were measured by Li and Sirkar [43]. They determined that $a_{\max} = 610$ nm, 330 nm, and 240 nm in the HVHP, GVHP and VVHP membranes, respectively. This demonstrates the problem that while the variation in pore size typically has a negligible effect on the transport properties provided that the liquid entry pressure is not exceeded, it may have a significant impact on the performance of the pressure-retarded process. Not only will the leakage through the larger pores result in leakage of solute into the permeate, but it also means that flow in the penetrated pores will be dominated by mechanically driven viscous flow from the permeate back into the feed, resulting in loss of both the desired permeate and of mechanical power.

We identify three parameters of particular importance when designing membranes for PRMD: the mean pore radius $\bar{a}$, the maximum pore radius $a_{\max}$, and the solid-liquid contact angle θ . We find that $a_{\max}$ is 3 to 5 times the mean pore radius in the XXHP membranes, and their solid-liquid contact angle is approximately 111°. A narrower pore size distribution moves $a_{\max}$ closer to $\bar{a}$, and the membrane surface can be modified to increase the contact angle. Both of these modifications increase the liquid entry pressure and thus allow the system to operate at a larger pressure difference. In Fig. 7, we have predicted the mechanical power density by use of Eq. (8), with the pressure difference in Eq. (11) and the resulting flux density J_w predicted by means of the method presented in [3]. We use the parameters of the VVHP membrane, in a case where the two liquid temperatures are 328 K and 283 K, with the equivalent of a 1 mol/kg NaCl solution on the feed side. The result is an optimized mechanical power density that depends on the pore size distribution.

We have introduced hypothetical cases where $a_{\max}$ is a multiple of $\bar{a}$, with $a_{\max} = \bar{a}$ being the ideal case with an extremely narrow pore size distribution. We have also included cases where the surface is modified

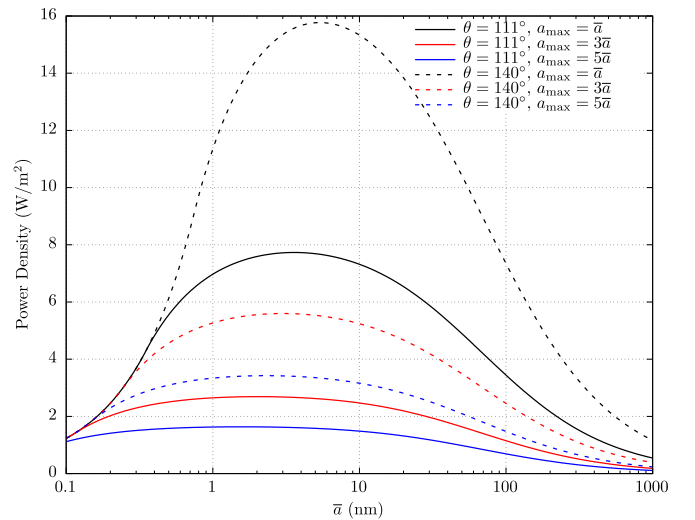


Fig. 7. The predicted mechanical power density of pressure-retarded membrane distillation as a function of the mean pore radius $\bar{a}$ for a process that operates close to the liquid entry pressure. The considered case has liquid temperatures 328 K and 283 K and salt content equivalent to 1 mol/kg NaCl on the feed side. The liquid entry pressure is determined by the maximum pore radius $a_{\max}$ and the solid-liquid contact angle θ .

in such a way that the contact angle is increased to 140° . For example, we find that the optimal mean pore radius in the ideal case $a_{\max} = \bar{a}$ and improved contact angle $\theta = 140^\circ$ is roughly 5.5 nm, with an ideal operating pressure difference of 190 bar. Provided that the membrane system can mechanically withstand this pressure difference, the mechanical power density would exceed 15 W/m^2 , and a production of nearly 3 kg/m^2 distilled water per hour would be maintained. By comparison, the VVHP membrane in its present state can achieve roughly 1 W/m^2 at a 2 bar pressure difference, with nearly 17 kg/m^2 water production per hour. If it would be possible to increase the contact angle of the VVHP membrane to 140° without changing the pore size distribution, one would be able to increase the operating pressure difference to 4.5 bar with a $<1\%$ reduction in water production, yet achieve more than twice the mechanical power density. For comparison, the benchmark minimum power density for pressure-retarded osmosis to be economically feasible was set to 5 W/m^2 [44], and in the PRMD case, power is only one out of two products.

4.3.2. Discussing the second law energy efficiency of membrane distillation

The entropy production (Eq. (6)) allows us to calculate the dissipated energy per unit time and the second law energy efficiency of the system. If the temperature of the environment is equal to T_1 , then the dissipated energy per unit time and surface area is $T_1\sigma$, and thus the dissipated energy per mole of distilled water produced is $w_{\text{lost}} = T_1\sigma/J_w$ [45]. The minimal work required to remove 1 M of water reversibly from the feed solution to the permeate reservoir is $w_{\text{id}} = -\Delta_{1,2}\mu_w$. The input energy to drive the process is that which maintains the temperature difference $\Delta_{1,2}T$. The second law efficiency of the process is the ratio of the minimal work required to the actual work consumed $w_{\text{tot}} = w_{\text{id}} + w_{\text{lost}}$,

$$\eta_{\text{II}} = 1 - \frac{w_{\text{lost}}}{w_{\text{tot}}} \quad (19)$$

The assessment of this quantity reveals a weakness of the PRMD process; thermal conduction across the membrane system is dominated by the relatively low thermal resistance of the solid membrane material. Reducing pore sizes to the ranges required to withstand the elevated pressures lowers the water flux, while the overall thermal conductance of the system remains relatively unchanged. This means that w_{lost} increases with reduction in pore size at a much greater rate than w_{tot} increases with the permissible pressure difference. Since the thermal losses are proportional to the square of the temperature difference (see the first term on the right-hand-side of Eq. (6)), while the water flux driving force is proportional to the temperature difference, the energy efficiency decreases with increasing temperature difference.

Fig. 8 shows the second law efficiency of a representative system with the VVHP membrane at different temperature differences, keeping the mean temperature of the system the same in all cases. We have also plotted the following figure of merit

$$ZT = \left(1 + \frac{\Lambda_{\text{tot}} L_p}{V_w T \Delta T}\right)^{-1} \quad (20)$$

The figure emphasizes two important points discussed in the previous paragraph; membranes with smaller pores are less energy efficient, and the magnitude of the temperature difference has a significant impact on the second law efficiency. Provided that the structural and wetting properties of the membrane are the same, thermal losses are more problematic with smaller pores due to the reduction in the mass fluxes. This means w_{lost} increases while w_{min} remains the same. It is therefore more important for the PRMD process than the standard isobaric DCMD process that the solid part of the porous membrane has a large thermal resistance, or that membrane systems are otherwise designed in a way that increases the thermal resistance without sacrificing mass permeability.

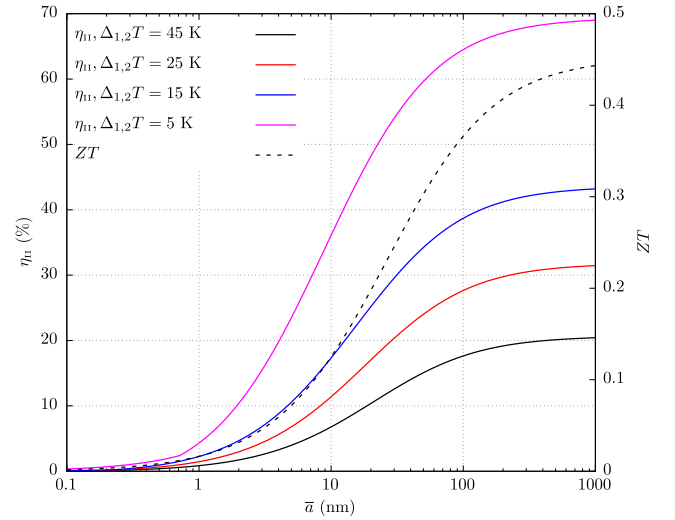


Fig. 8. The figure of merit, ZT , and the second law efficiency η_{II} of an example system with the Millipore Durapore VVHP membrane, at different temperature differences and a mean system temperature of 305.5 K.

5. Conclusion

The thermo-osmotic coefficient of a porous medium is the ratio of the mass flow rate and the temperature difference that drives the flow. An apparatus was designed and assembled for measuring the steady-state mass flux of water across a membrane system subject to simultaneous differences in temperature, composition, and pressure. The feed side of the apparatus was designed to withstand the corrosive properties of salt water at high temperature and pressure. The cell unit accommodates temperature measurements in the flow channels close to the membrane surface, giving a precise handle on the applied temperature difference.

Recommendations for data analysis and reduction for membrane distillation processes have been provided. Specifically, the apparatus measures mass rather than volume, eliminating the need to account for thermal expansion. The mass flux was obtained by measuring the rate of change in the fluid reservoir weights, and the reliability of these measurements was evaluated by confirming that the sum of the reservoir weights remained stable within the experimental uncertainty. We further recommended that the thermo-osmotic coefficient is reported by linear regression of the observed mass flux as a function of the applied temperature difference, keeping the mean temperature, compositions, and pressures constant; summarized by Eq. (13).

Applying the new apparatus with the recommended procedures, the thermo-osmotic coefficients of the Millipore DuraPore HVHP, GVHP and VVHP membranes were reported at several mean temperatures between 298 K and 328 K. Their dependence on the mean temperature was demonstrated to be in good agreement with the model developed in [3]. Due to the weak dependence of thermal polarization on the mean temperature, the temperature-dependence of the thermo-osmotic coefficient is accurately predicted by Eq. (5) with a temperature-dependent energy of activation given by Eq. (4), omitting the correction for temperature polarization (the last term) when this is not known a priori. The magnitude of the apparent energy of activation in Eq. (4) was found to be in agreement with experiments from similar systems reported in the literature for different membranes.

The measurements were repeated for the VVHP membrane with varying salt concentration on the feed side. The predicted effect of a 1 mol/kg salt concentration on the thermo-osmotic coefficient was a reduction of 2 %, which is too small to be detected in the experiments. Furthermore, the osmotic flux due to the chemical driving force imposed by the difference in composition was demonstrated to be small. This driving force is equivalent to a difference in mechanical pressure

provided that the liquid is unable to penetrate into the pore space. This fact shows great promise for the pressure-retarded membrane distillation (PRMD) process. Water can in theory be distilled against very large pressure differences without significant loss of distillation performance. The limiting factors in practice are the mechanical stability of the membranes and the minimal capillary pressure exerted by the largest pores.

An optimization scheme that respects the restriction imposed by the liquid entry pressure of the membrane was presented to predict the optimal pore radius to maximize the power density of the PRMD process. It was found that the optimal pore radii lie between 1 and 10 nm for membranes having characteristics similar to those of the XXHP series membranes, and that the power density is comparable to the lower end of viable values expected in pressure-retarded osmosis. The PRMD process benefits greatly from a narrower pore size distribution and surface modifications that increase the liquid-solid contact angle.

We conclude by pointing out the weakness revealed by calculating the entropy production and second law energy efficiency of the process. Smaller pores lead to smaller water fluxes, but the thermal conductivity of the membrane is dominated by thermal conduction through the solid material, which has a much higher thermal conductivity than the gas in the pores. Smaller fluxes means that it takes more time to produce a given amount of permeate, which means more time for exergy losses by thermal conduction across the membrane. We have shown that the PRMD process has a much lower second law efficiency than the isobaric process because for this reason. To improve the PRMD process, the thermal resistance of the membrane system should be increased. This can either be accomplished by developing a membrane with lower thermal conductivity, or by modifying the interfaces in such a way that the liquid-solid contact is minimized, as the liquid-solid contact has a much lower thermal resistance than the liquid-vapour interface. Increasing the contact angle can thus potentially increase the interfacial thermal resistance by several orders of magnitude.

CRediT authorship contribution statement

Kim R. Kristiansen: Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Øivind Wilhelmsen:** Conceptualization, Supervision, Writing – original draft, Writing – review & editing. **Signe Kjelstrup:** Conceptualization, Project administration, Supervision, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

M.H. Lundgreens Enkes Fond is thanked for helping to finance the pumps required to build the apparatus.

The authors thank Knut Jørgen Måløy at the Department of Physics/PoreLab of the University of Oslo for helping finance a large part of the investment in the new apparatus.

Thanks to Anders Jahres Fond For the Advancement of Science for their support in covering the costs of maintaining the apparatus and procuring membrane samples.

All authors are grateful to the Research Council of Norway through its Centers of Excellence funding scheme, Project No. 262644, PoreLab,

and to the Faculty of Natural Science, Norwegian University of Science and Technology, for a scholarship to K. R. K.

References

- [1] M.E. Findley, Vaporization through porous membranes, *Ind. & Eng. Chem. Process Des. Dev.* 6 (1967) 226–237.
- [2] M. Khayet, T. Matsuura, *Membrane Distillation: Principles and Applications*, Elsevier, 2011.
- [3] K.R. Kristiansen, Ø. Wilhelmsen, S. Kjelstrup, A systematic non-equilibrium thermodynamics approach for assessing transport mechanisms in membrane distillation, *Desalination* 567 (2023) 116927.
- [4] N. Kuipers, J.H. Hanemaaijer, H. Brouwer, J. van Medevoort, A. Jansen, F. Altena, P. van der Vleuten, H. Bak, Simultaneous production of high-quality water and electrical power from aqueous feedstock's and waste heat by high-pressure membrane distillation, *Desal. Water Treat.* 55 (2014) 2766–2776.
- [5] W.A. Phillip, Thermal-energy conversion: under pressure, *Nat. Energy* 1 (2016) 16101.
- [6] K.P. Straub, N.Y. Yip, S. Lin, J. Lee, M. Elimelech, Harvesting low-grade heat energy using thermo-osmotic vapour transport through nanoporous membranes, *Nat. Energy* 1 (2016) 16090.
- [7] A.P. Straub, M. Elimelech, Energy efficiency and performance limiting effects in thermo-osmotic energy conversion from low-grade heat, *Environ. Sci. Technol.* 51 (2017) 12925.
- [8] K. Park, D.Y. Kim, D.R. Yang, Theoretical analysis of pressure retarded membrane distillation (prmd) process for simultaneous production of water and electricity, *Ind. Eng. Chem. Res.* 56 (2017) 14888–14901.
- [9] Z. Yuan, L. Wei, J.D. Afroz, K. Goh, Y. Chen, Y. Yu, Q. She, Y. Chen, Pressure-retarded membrane distillation for low-grade heat recovery: the critical roles of pressure-induced membrane deformation, *J. Membr. Sci.* 579 (2019) 90–101.
- [10] Z. Yuan, Y. Yu, L. Wei, X. Sui, Q. She, Y. Chen, Pressure-retarded membrane distillation for simultaneous hypersaline brine desalination and low-grade heat harvesting, *J. Membr. Sci.* 597 (2020) 117765.
- [11] H. Cao, Y. Mao, W. Wang, Y. Gao, M. Zhang, X. Zhao, J. Sun, Z. Song, Thermal-exergy efficiency trade-off optimization of pressure retarded membrane distillation based on TOPSIS model, *Desalination* 523 (2022) 115446.
- [12] M.S. Lee, J.W. Chang, K. Park, D.R. Yang, Energetic and exergetic analyses of a closed-loop pressure retarded membrane distillation (PRMD) for low-grade thermal energy utilization and freshwater production, *Desalination* 534 (2022) 115799.
- [13] L. Keulen, L.V. van der Ham, N.J.M. Kuipers, J.H. Hanemaaijer, T.J.H. Vlugt, S. Kjelstrup, Membrane distillation against a pressure difference, *J. Membr. Sci.* 524 (2017) 151–162.
- [14] R. Zhao, Z. Zhang, W. Liu, Z. Liu, Harvesting net power and desalinating water by pressure-retarded membrane distillation, *Sci. China Technol. Sci.* 65 (2022) 214–220.
- [15] F. Mahmoudi, A. Date, A. Akbarzadeh, Further investigation of simultaneous fresh water production and power generation concept by permeate gap membrane distillation system, *J. Membr. Sci.* 572 (2019) 230–245.
- [16] F. Mahmoudi, A. Date, A. Akbarzadeh, Examining the commercially available hydrophobic membranes in combined desalination and power generation through permeate gap membrane distillation, *Desalination* 474 (2020) 114149.
- [17] M.T. Rauter, S.K. Schnell, S. Kjelstrup, Cassie-Baxter and Wenzel states and the effect of interfaces on transport properties across membranes, *J. Phys. Chem. B* 125 (2021) 12730–12740.
- [18] M.C. García-Payo, M.A. Izquierdo-Gil, Thermal resistance technique for measuring the thermal conductivity of thin microporous membranes, *J. Phys. D* 37 (2004) 3008.
- [19] M.C. García-Payo, M.A. Izquierdo-Gil, C. Fernández-Pineda, Wetting study of hydrophobic membranes via liquid entry pressure measurements with aqueous alcohol solutions, *J. Colloid Interface Sci.* 15 (2000) 420–431.
- [20] M. Khayet, M.P. Godino, J.I. Mengual, Modelling transport mechanism through a porous partition, *J. Non-Equilib. Thermodyn.* 29 (2004) 279–299.
- [21] J. Phattaranawik, R. Jiraratananon, Effect of pore size distribution and air flux on mass transport in direct contact membrane distillation, *J. Membr. Sci.* 215 (2003) 75–85.
- [22] M. Khayet, A.O. Imdakm, T. Matsuura, Monte Carlo simulation and experimental heat and mass transfer in direct contact membrane distillation, *Int. J. Heat Mass Transf.* 53 (2010) 1249–1259.
- [23] R.W. Schofield, A.G. Fane, C.J.D. Fell, Gas and vapour transport through microporous membranes. II. membrane distillation, *J. Membr. Sci.* 53 (1990) 173–185.
- [24] A. El-Abbassi, A. Hafidi, M.C. García-Payo, M. Khayet, Concentration of olive mill wastewater by membrane distillation for polyphenols recovery, *Desalination* 245 (2009) 670–674.
- [25] M.S. El-Bourawi, Z. Ding, R. Ma, M. Khayet, A framework for better understanding membrane distillation separation process, *J. Membr. Sci.* 285 (2006) 4–29.
- [26] M. Gryta, Osmotic md and other membrane distillation variants, *J. Membr. Sci.* 246 (1996) 145–156.
- [27] K.W. Lawson, D.R. Lloyd, Review: membrane distillation, *J. Membr. Sci.* 124 (1997) 1–25.
- [28] E. Curcio, E. Drioli, Membrane distillation and related operations: a review, *Sep. Purif. Rev.* 34 (2005) 35–86.

- [29] J.P.G. Villaluenga, S. Kjelstrup, A non-equilibrium thermodynamics model of multicomponent mass and heat transport in pervaporation processes, *J. Non-Equilib. Thermodyn.* 37 (2012) 353–376.
- [30] V.M. Barragán, S. Kjelstrup, Thermo-osmosis in membrane systems: a review, *J. Non-Equilib. Thermodyn.* 42 (2017) 217–236.
- [31] K. Smolders, A.C.M. Franken, Terminology for membrane distillation, *Desalination* 72 (1989) 249–262.
- [32] K. Gerstandt, K.-V. Peinemann, S.E. Skilhagen, T. Thorsen, T. Holt, Membrane processes in energy supply for an osmotic power plant, *Desalination* 224 (2008) 64–70.
- [33] T. Thorsen, T. Holt, The potential for power production from salinity gradients by pressure retarded osmosis, *J. Membr. Sci.* 335 (2009) 103–110.
- [34] L. Nilsson, S. Stenström, Gas diffusion through sheets of porous media, *Chem. Eng. Sci.* 50 (1994) 361–371.
- [35] V.D. Alves, I.M. Coelho, Effect of membrane characteristics on mass and heat transfer in the osmotic evaporation process, *J. Membr. Sci.* 228 (2004) 159–167.
- [36] S. Kim, M.M. Mench, Investigation of temperature-driven water transport in polymer electrolyte fuel cell: phase-change-induced flow, *J. Electrochem.* 156 (2009) 353–362.
- [37] K.R. Kristiansen, S. Kjelstrup, Particle flow through a hydrophobic nanopore: effect of long-ranged wall-fluid repulsion on transport coefficients, *Phys. Fluids* 33 (2021) 102001.
- [38] K. Bozkurt, L. Akyalçın, S. Kjelstrup, The thermal diffusion coefficient of membrane-electrode assemblies relevant to polymer electrolyte membrane fuel cells, *Int. J. Hydrog. Energy* 48 (2023) 1501–1513.
- [39] I.N. Tsimpanogiannis, O.A. Moulton, L.F.M. Franco, M.B. de M. Spera, M. Erdős, I. G. Economou, Self-diffusion coefficient of bulk and confined water: a critical review of classical molecular simulation studies, *Mol. Simul.* 45 (2019) 425–453.
- [40] A.W. Knight, N.G. Kalugin, E. Coker, A.G. Ilgen, Water properties under nano-scale confinement, *Sci. Rep.* 9 (2019) 8246.
- [41] W.M. Haynes, *CRC Handbook of Chemistry and Physics*, 96th edition, CRC Press, 2015.
- [42] M.P. Godino, L. Peña, C. Rincón, J.I. Mengual, Water production from brines by membrane distillation, *Desalination* 108 (1996) 91–97.
- [43] L. Li, K.K. Sirkar, Influence of microporous membrane properties on the desalination performance in direct contact membrane distillation, *J. Membr. Sci.* 513 (2016) 280–293.
- [44] S.E. Skilhagen, Osmotic power – a new, renewable energy source, *Desalin. Water Treat.* 15 (2010) 271–278.
- [45] S. Kjelstrup, D. Bedeaux, E. Johannessen, J. Gross, *Non-Equilibrium Thermodynamics for Engineers*, World Scientific, 2010.